Computer Science & IT



Fundamental of C Language

Structure & Union

Lecture-02



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Recap of Previous Lecture



- Topic — string
- Topic — string declaration & initialization
- Topic — string print
- Topic — strlen() function
- Topic

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Topics to be Covered



Topic Structure

Topic Structure Declaration

Topic Structure Member

Topic "." Dot operator.

Topic

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Structure

Data & Data type

* Structure gives the power to define user its own Data type
* Structure and union is in user defined data type



Structure

Student information we create Student data type

```
struct student {
```

struct → Keyword

student → Name of data type

```
    char name[20];
    int Roll_No;
};
```

char name[20]; → Member of Structure

int Roll_No; → Member of Structure

} ; ← Semicolon

Slide



Structure can be declared as global as well as local

Declaration of structure

variable

int a;

Struct student S;

S is a structure variable of type student



Initialization of structure

variable

```
struct student S = { "Ravi", 120};
```



Structure

To access member of structure using structure variable we use dot (".") operator

```
struct student S = {"Ravi", 120};

printf ("%s", S.name);

printf ("%d", S.Roll_No);
```



Structure

```
# include<stdio.h>
struct test {
        int i;
        char ch;
} t1;
```

t_1 is a structure variable

```
int main() {
    struct test t = {15, 'A'};
    printf("%d", t.i);
    printf("%c", t.ch);
    t.i = t.i+10;
    t.ch = t.ch+1;
    printf("%d", t.i); // 25
    printf("%c", t.ch);
    return 0; }
```

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Structure

```
# include<stdio.h>
struct test {
    int i;
    char ch;
} t1;
```

Memory Allocation



4 Bytes 1 Byte

Memory allocated for every member of Structure whenever structure variable is created.



include<stdio.h>

Union

union test {

int i; // 4Byte

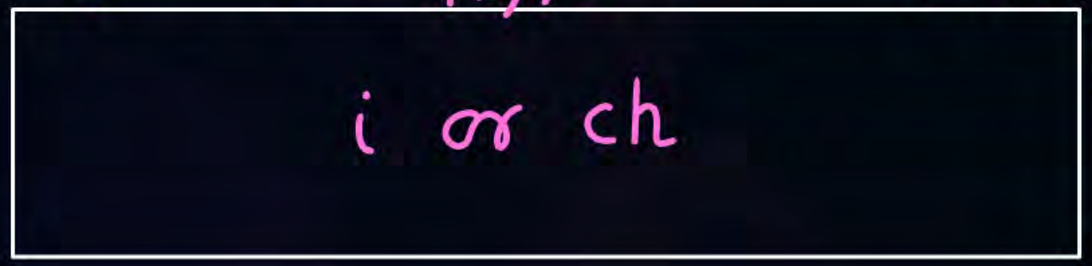
char ch; // 1Byte

};

Instead of structure union is written

Memory allocation max(Memory Needed for Members)

<4Bytes>




```
int main(){
  union test t;
  t.i = 15;
  printf("%d", t.i);
  t.ch = 'a';
  printf("%c", t.ch);
  return 0;
}
```

Slide



Union

At one time only 1 Member can be stored.

Either integer or character will be stored in same memory Location.



2 mins Summary

Topic — Structure declaration & initialization

Topic — Structure Member

Topic — Dot "." operator

Topic — Union

Topic

THANK - YOU